LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (AAT) (31)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Various types of common hand-held tools, power tools and their correct usage
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Various types of common hand-held tools, power tools and their correct usage
5. Specific Objectives	31.1.2.1 Orientation of trainee in various types of AE skills

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration and handling of common hand-held and power tools with emphasis on safety precautions and instructor supervision
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary categories of tools used in aircraft maintenance?
(b) Why is correct tool selection critical for aircraft structural integrity?
(c) Can you list three safety precautions when using power tools?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Classification of hand-held tools • Common screwdrivers and wrenches • Pliers and cutting tools • Measuring instruments • Selection criteria for specific tasks	L & Ds	PPT, IB	15 min
(b)	• Types of power tools • Drills and impact drivers • Grinders and sanders • Safety features of power tools • Maintenance of power tools	L & Ds	PPT, PDF	20 min
(c)	• Correct usage techniques • Storage and organization • Troubleshooting common issues • Regulatory compliance • Best practices for tool care	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a manual and a power tool in terms of application.
(b) Describe the correct procedure for storing sensitive measuring instruments.
(c) What are the immediate steps to take if a power tool malfunctions during use?

12. Home Assignment and Follow Up	Prepare a brief report on the specific hand-held tools required for the next lesson on removing rounded screws, referencing the tool list in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Easy out / Chisel out of rounded screws on aircraft structure, Stop drilling of cracks	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min